7	**(a)**	A spring that has an unstretched length of 0.500 m is attached to a fixed point. A mass of 0.400 kg is attached to the spring and gently lowered until equilibrium is reached. The spring has then stretched elastically by a distance of 0.150 m. Calculate, for the stretching of the spring	
		(i) the loss in gravitational potential energy of the mass,	
		loss = ……………… J	[1]
		loss $= mgh = 0.400 \times 9.81 \times 0.150 = 0.589\,J$	
		(ii) the elastic potential energy gained by the spring.	
		gain = ……………… J	[1]
		Gain in EPE $= \frac{1}{2}Fx = \frac{1}{2}mgx = \frac{1}{2} \times 0.400 \times 9.81 \times 0.150 = 0.294\,J$	
	(b)	Explain why the two answers to **(a)** are different	
		………………………………………………………………………………………………	

[Turn over

………

…… [2]

When the mass is lowered gently, there is a resistive or supporting force that prevents the mass from accelerating. [B1]
As a result, part of the loss in gravitational potential energy is into the work done against the supporting force [B1]
And the gain in elastic potential energy is less than the loss in gravitational potential energy.

(c) The load on the spring is now set into simple harmonic motion of amplitude 0.150 m.

Show that the angular frequency of the oscillation is 8.09 rad s^{-1}.

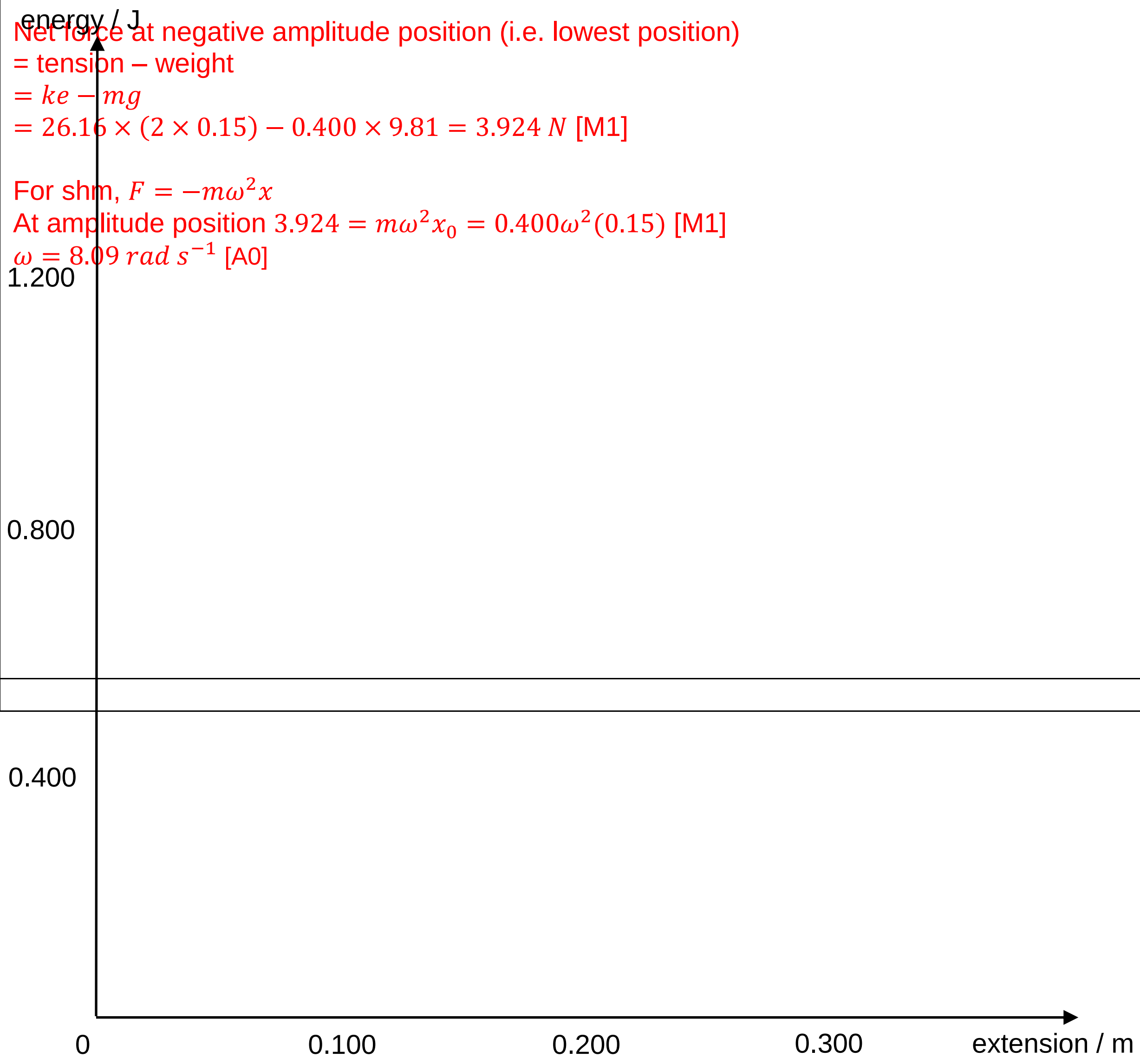
[3]

Spring constant $= \frac{F}{e} = \frac{mg}{e} = \frac{0.400 \times 9.81}{0.15} = 26.16\ N\ m^{-1}$ [C1]

Net force at negative amplitude position (i.e. lowest position)
= tension – weight
$= ke - mg$
$= 26.16 \times (2 \times 0.15) - 0.400 \times 9.81 = 3.924\ N$ [M1]

For shm, $F = -m\omega^2 x$
At amplitude position $3.924 = m\omega^2 x_0 = 0.400\omega^2(0.15)$ [M1]
$\omega = 8.09\ rad\ s^{-1}$ [A0]

(d)



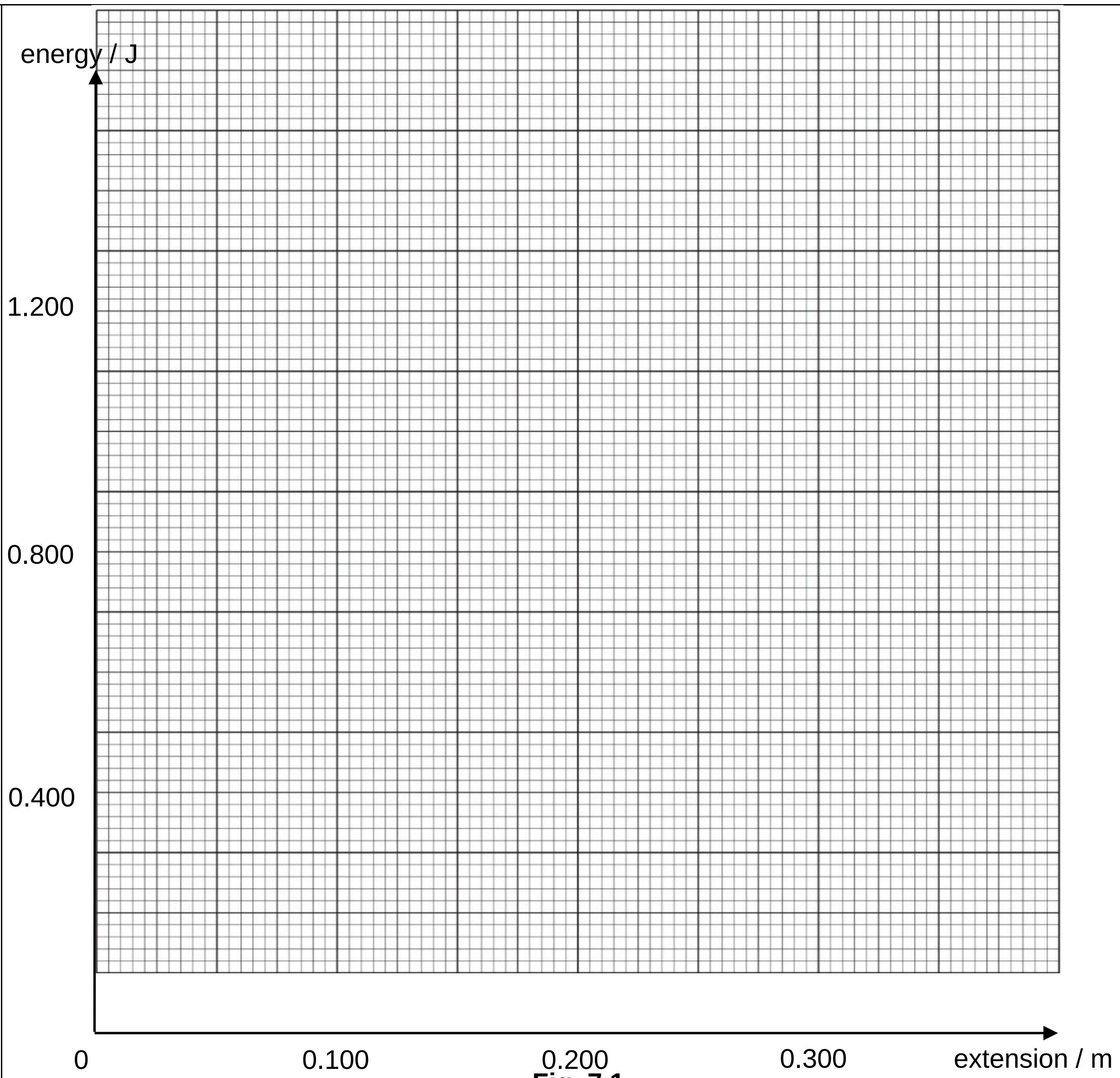


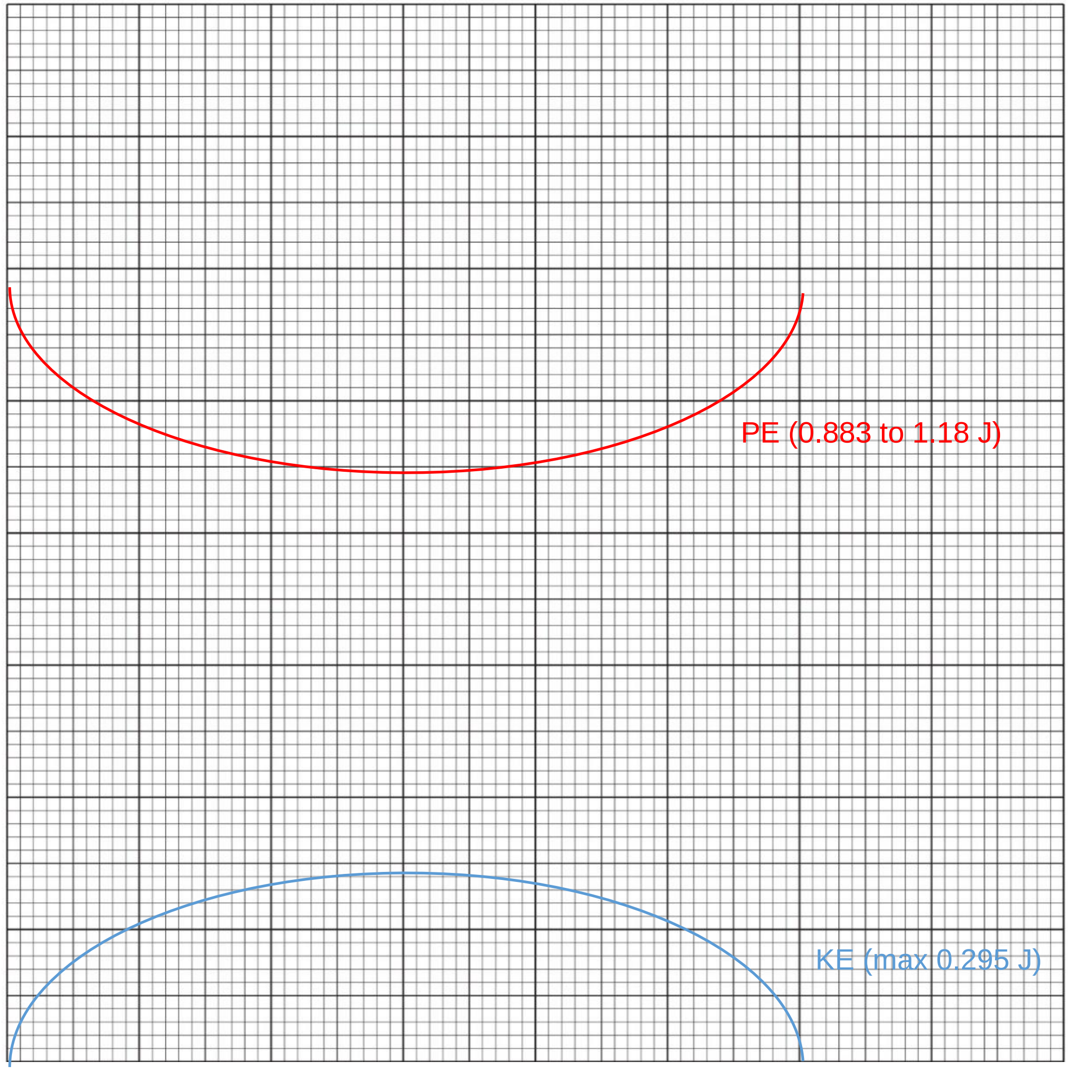
Fig. 7.1

The gravitational potential energy of the mass is 0 J when the spring is fully extended to the lowest point of the oscillation of the mass.

On Fig. 7.1, sketch the variation with extension of the spring, the kinetic energy and total potential energy for the oscillating mass. Label the graph for kinetic energy T and the graph for total potential energy V.

[3]

(d)



KE graph (correct shape and values with reference to equilibrium) [B1]
GPE graph (correct shape and relative values at amplitude and equilibrium) [B1]
GPE graph (correct absolute values) [B1]

			[Total: 10]

8 In a refrigeration system, a fluid known as a refrigerant undergoes alternating changes in its phase